Multi-Step Weather Forecasting — Final Report

LSTM · GRU · Bi-LSTM on the Jena Climate Dataset

1. Executive Summary

We built and compared classical and deep-learning models to forecast air temperature from historical weather observations. The best single-step model was **XGBoost** (RMSE **0.640 °C**, MAE **0.450 °C**, R^2 **0.9932**). Deep recurrent models were extended to multi-step forecasting (24h / 48h / 7-day) for realistic operational use, and SHAP analysis identified the dominant physical drivers.

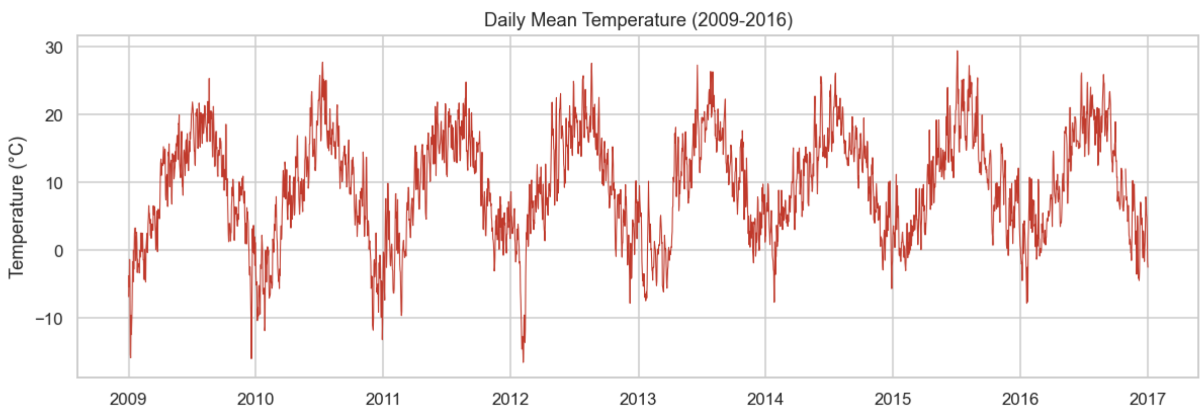
2. Business Problem

Accurate temperature forecasting drives decisions across **airlines, agriculture, energy, logistics, and smart cities** — from energy demand planning to crop protection and transport scheduling. The goal: predict future temperature from historical multivariate weather conditions.

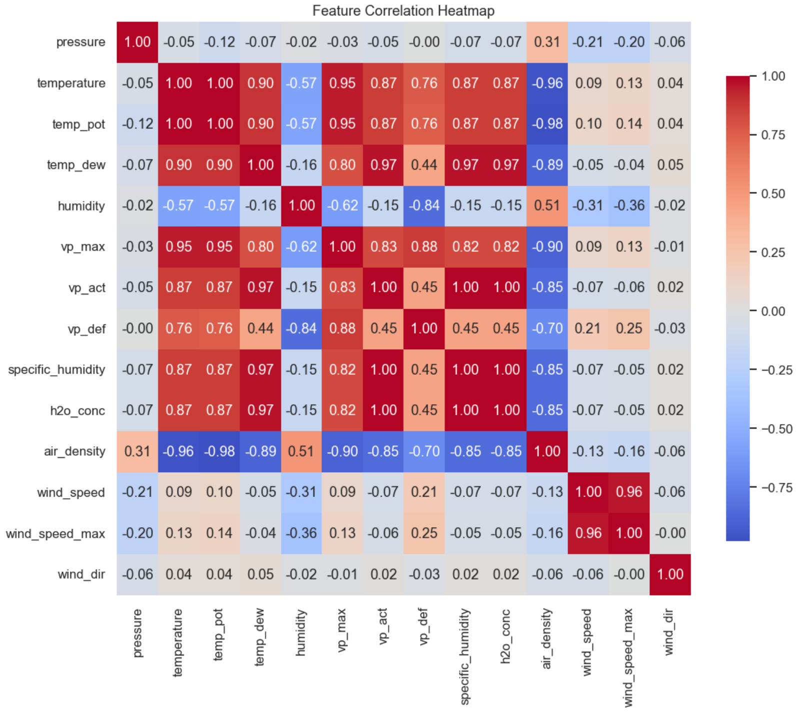
3. Data Understanding

- **Source:** Jena Climate 2009–2016 (Max Planck Institute).
- **Raw records:** 420,551 at 10-minute resolution (2009-01-01 00:10:00 → 2017-01-01 00:00:00).
- **Features:** 14 meteorological variables (temperature, pressure, humidity, dew point, vapor pressure, air density, wind speed/direction).
- **Resampling:** hourly (≈70k rows) for tractable, leakage-free modeling.
- **Data quality:** missing values = none; duplicate timestamps = 327 (dropped); impossible wind readings = 38 (treated as missing, filled).

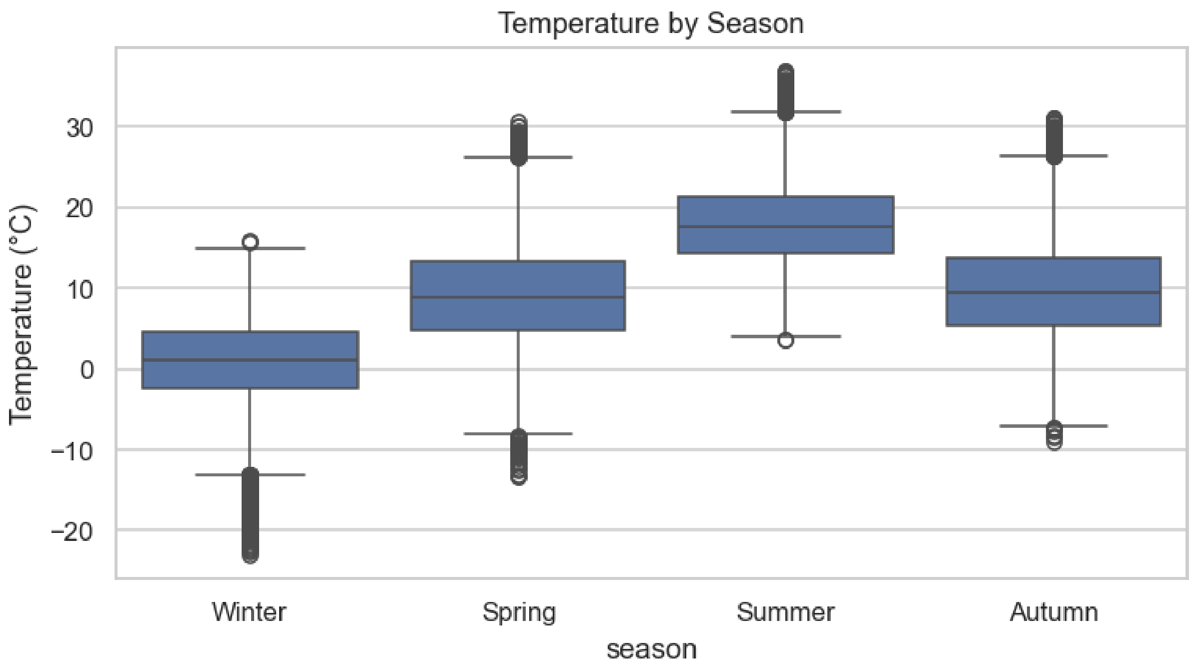
4. EDA Findings



Daily mean temperature 2009–2016



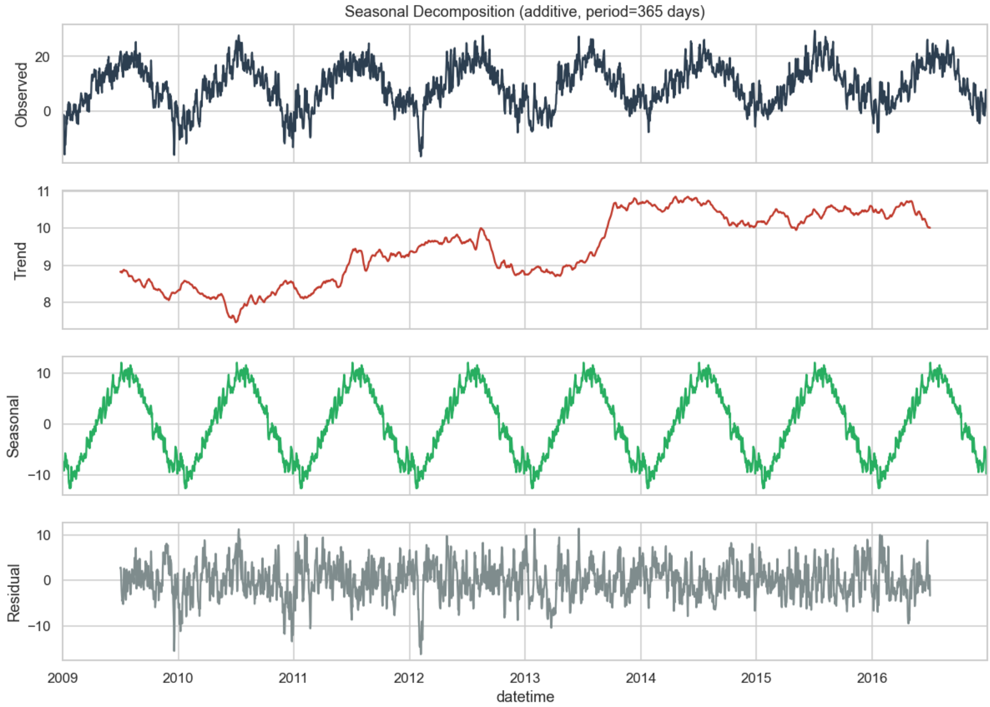
Feature correlation heatmap



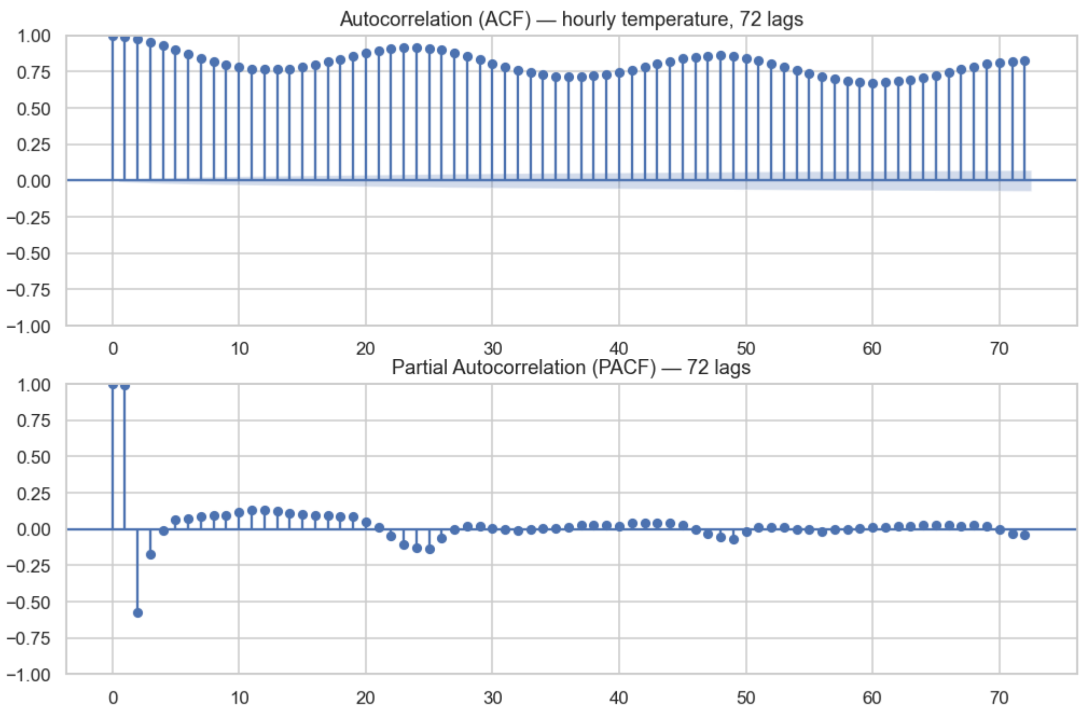
Temperature distribution by season

Key observations: strong annual seasonality (summer–winter swing of ~17 °C); temperature is highly correlated with dew point, vapor pressure and (negatively) air density; humidity is inversely related to temperature; wind speed is right-skewed and mildly seasonal.

5. Time-Series Analysis



Additive seasonal decomposition



ACF / PACF of hourly temperature

- Long-term trend: **0.321 °C/year**.
- Summer mean **18.04 °C** vs winter **0.78 °C**.
- Autocorrelation: lag-1 = **0.993**, lag-24 = **0.918** — strong daily memory, which motivates sequence models.

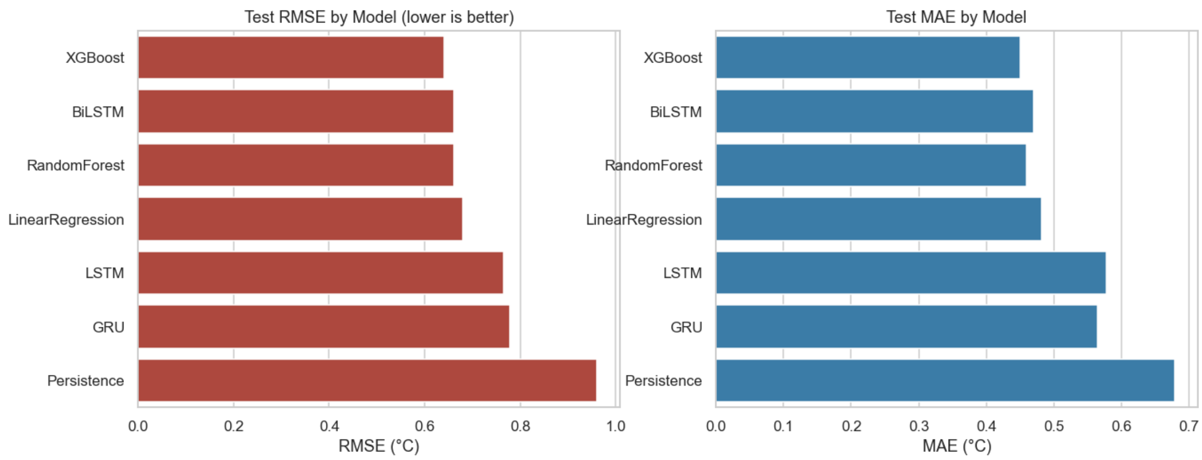
6. Modeling

Feature engineering: lag features (t-1, t-6, t-24, t-48), rolling mean/std (7h, 24h, 48h), calendar features and cyclic (sin/cos) encodings of hour, day-of-year and wind direction.

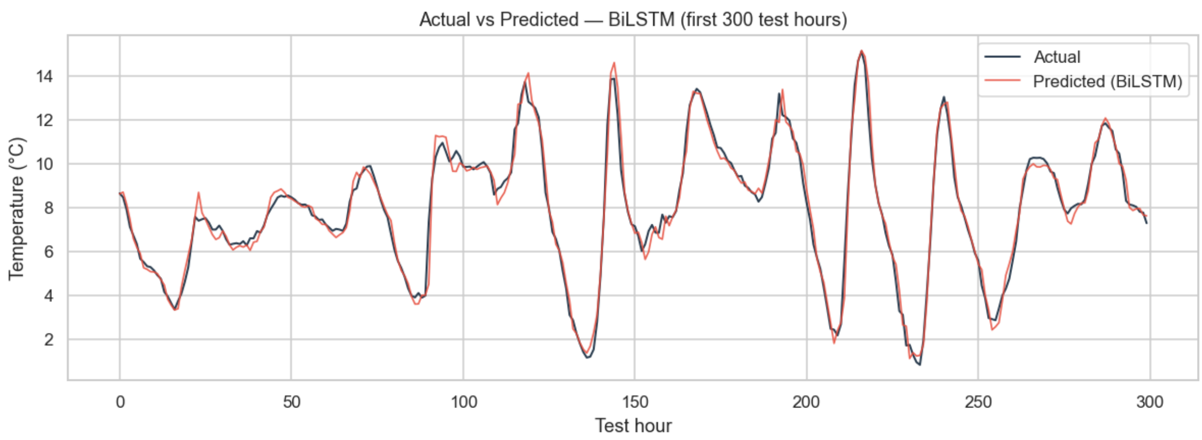
Models: Persistence baseline → Linear Regression → Random Forest → XGBoost → LSTM → GRU → Bi-LSTM. All evaluated on the same chronological 70/15/15 split with train-only scaling.

Single-Step Leaderboard (test set)

	MAE	RMSE	MAPE	R2
XGBoost	0.4502	0.6405	7.138	0.9932
BiLSTM	0.4697	0.6609	7.502	0.9928
RandomForest	0.4591	0.662	7.219	0.9928
LinearRegression	0.4824	0.6806	7.826	0.9924
LSTM	0.5779	0.7655	10.757	0.9903
GRU	0.5644	0.7789	8.278	0.99
Persistence	0.6788	0.9607	10.006	0.9848



RMSE / MAE by model



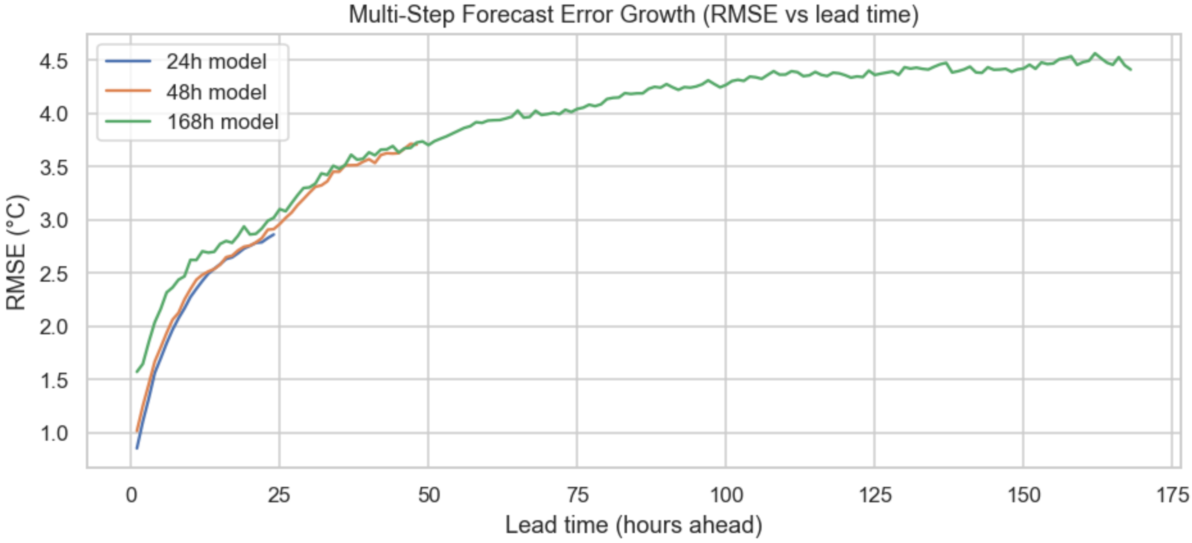
Actual vs predicted (best DL model)

7. Hyperparameter Tuning

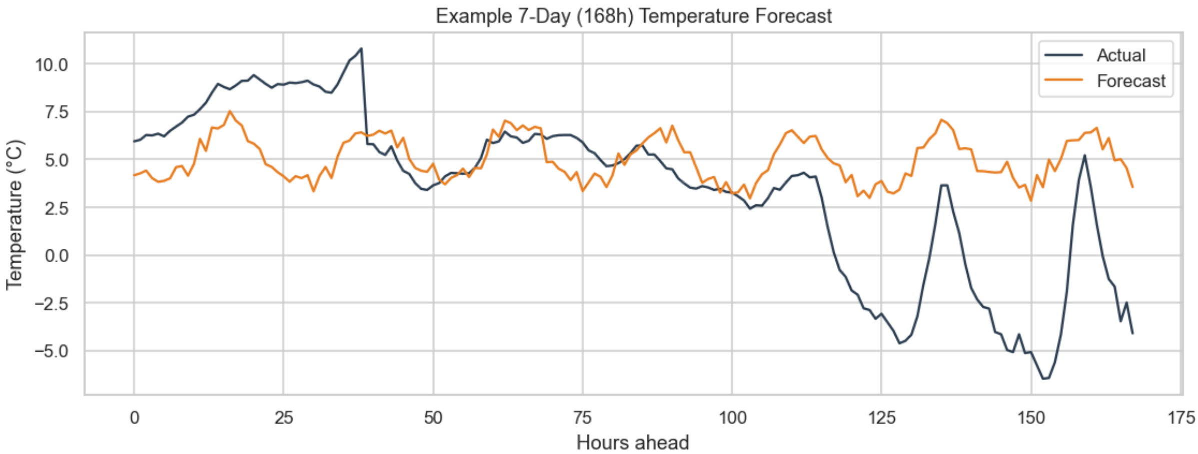
Random search over window size, hidden units, dropout, learning rate and batch size. **Best configuration:** `{"window": 24, "units": 128, "dropout": 0.1, "lr": 0.0005, "batch": 128, "arch": "BiLSTM"}` (val RMSE 0.6961).

8. Multi-Step Forecasting

	MAE	RMSE	MAPE	R2	per_step_rmse_first	per_step_rmse_last
24h	1.7743	2.3174	31.082	0.9112	0.8507	2.8597
48h	2.2636	2.949	37.352	0.8559	1.0137	3.7043
168h	3.0843	3.954	49.719	0.7403	1.5697	4.4093



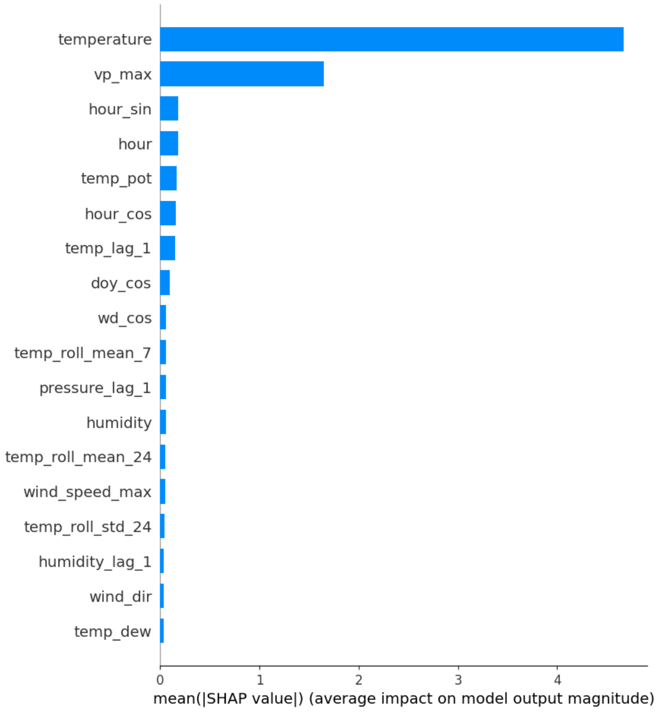
Error growth with lead time



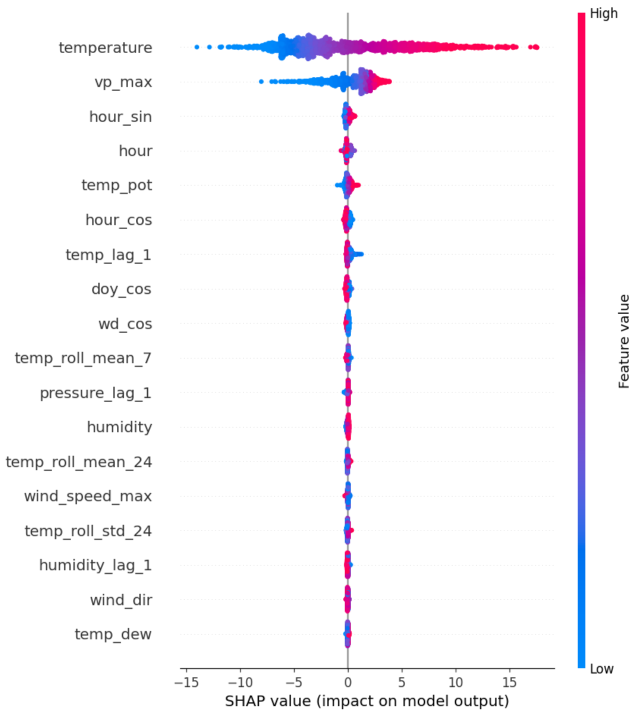
Example 7-day forecast

As expected, error grows with lead time; the 24h horizon is the most accurate and the 7-day horizon the hardest.

9. Explainability (SHAP)



Global SHAP feature importance



SHAP beeswarm (direction & magnitude)

Key physical drivers of the next-hour prediction (mean |SHAP|):

- `temp_lag_1`: 0.1465
- `pressure_lag_1`: 0.0557
- `humidity`: 0.0544
- `temp_roll_mean_24`: 0.0493
- `humidity_lag_1`: 0.0350
- `wind_speed`: 0.0254

- `pressure`: 0.0151

Recent-temperature lags dominate (thermal inertia), with humidity and pressure providing the meteorological context that the model uses to adjust the short-term trajectory.

10. Business Impact

- **Energy:** day-ahead temperature forecasts sharpen heating/cooling demand prediction and grid load balancing.
- **Agriculture:** frost and heat-stress early warning for crop protection and irrigation scheduling.
- **Transport & logistics:** proactive scheduling around adverse conditions, reducing delays.
- **Smart cities:** automated HVAC and public-service planning.

Generated from the project's saved artifacts in `outputs/`.